

Arle Lommel, PhD

MULTILINGUAL AI, LOCALIZATION, AND
RESEARCH STRATEGY LEADER

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EXECUTIVE PROFILE

Research leader specializing in multilingual AI, machine translation, localization quality, human-machine interaction, and the transformation of global content. Nearly three decades of experience spanning applied research, localization technology, international standards, quantitative analysis, cultural research, and business strategy.

Key Achievements

- ▶ **Vice President of Research at CSA Research** – develops computational methods ranging from deterministic algorithms and individual-level simulation to Bayesian modeling, ML, and generative AI. Leads a research agenda and team of analysts focusing on accessibility, localization, multilinguality, and multimedia, and cultural customization.
- ▶ **Developed applied research systems** spanning individual-level customer simulation, economic forecasting, Bayesian/Monte Carlo market modeling, and GenAI strategic intelligence to turn multilingual data into business decisions.
- ▶ **Creator and lead developer of Multidimensional Quality Metrics (MQM)**, now widely used across the localization industry and by the machine translation research community, including by Google Research and WMT.
- ▶ **Led development of core localization interoperability standards** including TMX, TBX, SRX (ISO 30042), GMX-V, enabling multilingual data to move freely between tools and systems. Co-editor of the W3C ITS 2.0 Recommendation.
- ▶ **Co-creator of the Metrix 2.0 capability-maturity model** used by language service providers to benchmark their performance and drive strategic business improvement.

PROFESSIONAL EXPERIENCE

1

CSA Research (Cambridge MA)

Senior Analyst (2016) • Director, Data Services (2020); VP of Research (2024)

**2016–
present**

- ▶ Designs and prototypes analytical systems using a variety of computational methods – deterministic algorithms, individual-level simulation, probabilistic modeling, etc. – to solve challenging research questions.
- ▶ Leads research strategy, methodology, and execution with a team of 3–4 senior analysts and support staff for an independent research organization with Global 1,000 and language services clients, focused on globalization, multilingual customer experience, and the value of language. Interfaces with CEOs of leading enterprises in the language sector.
- ▶ Sets and evolves the research agenda as technological and market conditions change, guiding the organization increasingly toward computational, AI-enabled, and just-in-time research methods rather than periodic analyst reports.
- ▶ Developed a GenAI strategic-intelligence system to combine client information with CSA Research's reports to generate customized analysis and strategic guidance, moving the company toward decision support for individual organizations.
- ▶ Visit arlelommel.github.io/projects/ for more on the systems I design and built to support CSA Research's analysis.

Key Research Topics and Skills

- | | | |
|---|--|------------------------------------|
| • Modeling international buyer behavior | • AI-assisted software development | • Stakeholder Management |
| • Survey research | • LLM-assisted information extraction | • Mentoring & career development |
| • Textual/corpus analysis | • Multilingual workflows and processes | • Multimodal AI application |
| | • Research team management | • Cultural research and adaptation |

2

MQM Council (Bloomington IN)

President

2025–
present

- ▶ Transitioned the MQM Council from an informal technical committee into a nonprofit, providing a durable institutional basis for governance, education, standards work, and continued development of MQM
- ▶ Guides the continued development of Multidimensional Quality Metrics (MQM), enabling organizations to define quality requirements, and evaluate human and machine translation to manage language-related risk more effectively.

3

German Research Center for Artificial Intelligence (DFKI)

Senior Consultant, Berlin Language Technology Lab

2012–
2015

- ▶ Led applied research projects at the intersection of machine translation, translation quality, multilingual information, interoperability, and language-technology standards that involved participants from around the world
- ▶ Contributed to standards and European Union research programs addressing translation quality, multilingual linked data, interoperable language technologies, and the machine-readable representation of multilingual information.
- ▶ Helped organize and lead the *Multilingual Web Workshop* series with leading researchers, standards specialists, and developers to address technical barriers to a multilingual web.

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Localization Industry Standards Association (LISA)

Director of Standards and Chair of OSCAR standards committee

1998–
2011

- ▶ Ran research publication, industry standards development, technical coordination, and localization-technology strategy
- ▶ Chaired the OSCAR standards committee, resulting in core localization standards: TMX, SRX, & TBX (ISO 30042).
- ▶ Served as liaison for international language-technology standardization with the ISO Technical Committee 37.
- ▶ Ran Executive Roundtable sessions that brought together leaders of enterprises and language services companies to derive shared solutions to industry challenges

EDUCATION

- ▶ PhD (2010) and MA (2005), Indiana University, Bloomington (2005)
- ▶ BA, Brigham Young University (1997)

LANGUAGES

- ▶ English – Native • Hungarian – Fluent • German – Conversational